

IV. RESULTS AND ANALYSIS

A. Experimental Setup

All results in this section use the leak-free, file-level/intra-file train/validation/test split described in Section III, in which flows from the same capture session are confined to a single split. This differs from the naive row-level random split used in the dataset's original publication, which permits flows from the same session to appear in both training and evaluation. Reinforcement learning (RL) policies and classifier-derived policies are evaluated inside ICS-NAD-Gym using a fixed evaluation seed, so that every arm — each of the ten supervised classifiers, the RecurrentPPO baseline, and the RecurrentPPO + LLM-ShAId arm — is scored against an identical sequence of held-out episodes, removing episode-sampling variance as a confound between arms. Classifier decision thresholds (Section IV-C) are tuned exclusively on the validation split and then held fixed for test-split evaluation. Reported test-split F_1 confidence intervals are episode-level bootstrap intervals (2000 resamples, 95% percentile interval) computed over 40 held-out episodes per arm.

B. Baseline Classifier Performance Under a Leak-Free Evaluation Protocol

Table I reports validation- and test-split F_1 for the ten supervised classifiers reproduced from the dataset's companion publication, trained and evaluated under the leak-free split. A pronounced generalization gap is observed for instance-based and margin-based classifiers: K-nearest neighbors (KNN), the support vector classifier (SVC), and the extreme learning machine (ELM) each achieve near-perfect validation F_1 (1.000) but collapse to $F_1 = 0.831$ on the held-out test split, a drop of 0.169. Gaussian naïve Bayes shows a comparable collapse ($0.898 \rightarrow 0.711$). By contrast, tree-ensemble methods (GBDT, Random Forest, AdaBoost, LightGBM, XGBoost) generalize cleanly across the session boundary; GBDT's test F_1 (0.985) exceeds its validation F_1 (0.894).

TABLE I
STATIC CLASSIFIER F_1 UNDER THE LEAK-FREE SPLIT

Model	Val. F_1	Test F_1	Δ
GBDT	0.894	0.985	+0.091
Random Forest	0.931	0.920	-0.011
AdaBoost	0.936	0.917	-0.019
LightGBM	0.922	0.917	-0.005
XGBoost	0.922	0.916	-0.006
ANN	0.930	0.886	-0.044
ELM	1.000	0.831	-0.169
KNN	1.000	0.831	-0.169
SVC	1.000	0.831	-0.169
Gaussian NB	0.898	0.711	-0.187

This asymmetry — ensemble methods generalizing cleanly, instance-based and margin-based methods overfitting to session-specific characteristics — is consistent with the split-leakage phenomenon documented for flow-based network intrusion detection datasets more broadly [1], and demonstrates that the effect is architecture-dependent rather than uniform across classifier families.

C. Classifiers as Sequential Policies: Threshold-Optimized Response Mapping

Static classification does not, by itself, determine operational value under an asymmetric response cost structure. Each classifier in Table I was therefore deployed inside ICS-NAD-Gym as a sequential policy over the three-action space {ALLOW, ALERT, BLOCK}, using `predict_proba()` output thresholded at two learned cut-points $t_{\text{low}} < t_{\text{high}}$ (grid-searched over {0.20, 0.35, 0.50, 0.65, 0.80} on the validation split to maximize mean episodic reward, then fixed for test evaluation) rather than the naive binary mapping `predict()` $\rightarrow$ {ALLOW, BLOCK} used in prior work. Table II reports the tuned thresholds and resulting test-split reward and F_1 (with bootstrap 95% CI), alongside the binary-mapping reference.

TABLE II
THRESHOLD-TUNED VS. BINARY CLASSIFIER POLICIES (TEST SPLIT)

Model	t_low / t_high	Reward	F1 [95% CI]	Binary F1
ANN	0.35 / 0.80	2678.9	0.937 [0.883, 0.980]	0.937
Random Forest	0.35 / 0.50	2609.8	0.936 [0.883, 0.978]	0.936
ELM	0.50 / 0.65	2585.7	0.933 [0.878, 0.978]	0.933
LightGBM	0.35 / 0.50	2414.2	0.931 [0.876, 0.975]	0.931
XGBoost	0.20 / 0.35	2383.7	0.928 [0.873, 0.972]	0.927
GBDT	0.20 / 0.35	2358.5	0.924 [0.869, 0.970]	0.941
KNN	0.65 / 0.80	1243.5	0.869 [0.773, 0.939]	0.871
AdaBoost	0.50 / 0.65	3279.3	0.867 [0.796, 0.927]	0.867
SVC	0.20 / 0.35	918.6	0.848 [0.749, 0.921]	0.847
Gaussian NB	0.20 / 0.35	128.2	0.811 [0.723, 0.889]	0.811

Two observations follow. First, reward and F_1 do not always co-vary: AdaBoost attains the highest reward of any classifier (3279.3) at unchanged F_1 (0.867 under both mappings), because the ALERT action allows a subset of its false-positive-costly BLOCK decisions to be converted into lower-cost alerts without altering detection accuracy — a distinction invisible to an F_1 -only evaluation. Second, threshold tuning improves or matches binary-mapping reward for nine of ten classifiers; GBDT is the exception, trading a modest F_1 reduction (0.941 $\rightarrow$ 0.924) for higher reward (2142.3 $\rightarrow$ 2358.5), again indicating that the reward-optimal operating point does not coincide with the F_1 -optimal one.

D. Reinforcement Learning Baseline

RecurrentPPO (sb3-contrib, MlpLstmPolicy), trained for 2×10^6 environment steps on the base (unshaped) reward, attains test-split mean reward 2096.0 ($F_1 = 0.818$, precision = 0.886, recall = 0.760), exceeding the weakest classifier policies

(Gaussian NB: reward 128.2) and falling within the range of the mid-tier threshold-tuned classifiers (Table II), while additionally providing stateful quarantine behavior and a detection-latency-aware reward component that no static classifier formulation expresses.

E. Effect of LLM-Guided Reward Shaping: A Five-Seed Ablation

The central ablation isolates the contribution of LLM-ShAId, a potential-based reward-shaping term derived from LLM-scored flow archetypes (Section III). RecurrentPPO was trained to a matched budget of 2×10^6 steps under five random seeds (0–4), with identical hyperparameters and per-environment seeding between the shaped and unshaped arm of each seed; the shaping term is the only difference between arms. Because potential-based shaping of the form $F(s, s') = \gamma\Phi(s') - \Phi(s)$ is provably policy-invariant [2], any measured difference between arms is attributable to the guidance quality of the shaping signal during learning rather than to a changed optimum.

TABLE III
FIVE-SEED ABLATION: LLM-GUIDED REWARD SHAPING (TEST SPLIT)

Seed	No-Shaping F1 [95% CI]	Shaping F1 [95% CI]	$\Delta F1$
0	0.818 [0.714, 0.893]	0.853 [0.766, 0.918]	+0.035
1	0.781 [0.694, 0.859]	0.886 [0.820, 0.937]	+0.104
2	0.895 [0.827, 0.946]	0.902 [0.838, 0.953]	+0.007
3	0.883 [0.812, 0.936]	0.886 [0.815, 0.939]	+0.003
4	0.869 [0.798, 0.925]	0.835 [0.740, 0.902]	−0.034
Mean $\pm$ SD	0.849 $\pm$ 0.048	0.872 $\pm$ 0.027	+0.023

Across the five seeds, mean test F_1 is 0.849 ± 0.048 (no shaping) versus 0.872 ± 0.027 (shaping); mean test reward is 2828.3 ± 1415.7 versus 3309.6 ± 753.6 , respectively. The paired difference (shaping – no-shaping) has mean +0.023 with a 95% t -based confidence interval of $[-0.041, 0.087]$, which contains

zero. A Wilcoxon signed-rank test gives $W = 3.0$, $p = 0.3125$; a paired one-sample t -test gives $t(4) = 1.007$, $p = 0.371$. Neither test rejects the null hypothesis of no effect at $\alpha = 0.05$.

The sign of the effect is non-negative in four of five seeds (three positive, one effectively flat at +0.003), with one reversal (seed 4: $\Delta F_1 = -0.034$, reward 3753.1 $\rightarrow$ 2509.9). A preliminary analysis restricted to the first three seeds yielded a substantially larger mean effect (+0.049, all three seeds positive); the full five-seed result revises this estimate downward, illustrating the risk of treating a small- n interim result as confirmatory. A retrospective power calculation using the observed effect size ($d = 0.45$, from mean/SD of the five paired differences) indicates that approximately $n = 39$ seeds would be required to detect an effect of this magnitude at $\alpha = 0.05$, power = 0.8, under a paired t -test — substantially more than the five seeds evaluated here. Notably, the shaping condition exhibits lower cross-seed variance in both F_1 (SD 0.027 vs. 0.048) and reward (SD 753.6 vs. 1415.7) than the unshaped condition, suggesting a possible variance-reduction effect independent of, and not captured by, the mean-shift significance tests above; this observation is not statistically tested here and is noted as a candidate hypothesis for follow-up work rather than a confirmed finding.

F. Discussion

Three results are robust to the sample-size limitations of Section IV-E and do not depend on the shaping ablation reaching statistical significance. First, the leak-free split reveals a real, architecture-dependent generalization gap (Section IV-B) invisible under the dataset's original evaluation methodology, consistent with independently reported concerns about split leakage in flow-based network intrusion detection evaluation [1]. Second, F_1 -score does not reliably predict operational value under an asymmetric cost structure (Section IV-C); this is consistent with independently reported concerns that uniform accuracy/ F_1 metrics overlook the asymmetric cost of missed detections in security operations [3]. Third, probability-threshold tuning recovers most of this gap for static classifiers without requiring a full RL formulation, though the RL policy additionally provides stateful response behavior (quarantine, detection-latency sensitivity) that a per-flow classifier cannot express by construction. The LLM-guided shaping result (Section IV-E) is reported as suggestive rather than conclusive: the effect is directionally favorable in four of five seeds and the observed variance reduction is a plausible secondary benefit, but the paired confidence interval does not exclude zero at the evaluated sample size.

[1] Network Intrusion Datasets: A Survey, Limitations, and Recommendations, arXiv:2502.06688, 2026.

[2] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in *Proc. ICML*, 1999.

[3] Cost-Sensitive Threshold Optimization for Network Intrusion Detection: A Per-Class Approach with XGBoost, *Electronics*, 2026, doi:10.3390/electronics15071542.